The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

Richard Sherman
System Architecture Invariant Manifest

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1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF). The ultimate objective of the system is the minimization of coordinate uncertainty to realize Mesarovic's Attainable Space (V'), culminating in physical vehicle extraction.

2 The 30-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 30}$:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{emp (16)}} \mid \boldsymbol{g}_i^{\text{spatial (2)}} \mid \boldsymbol{p}_i^{\text{ppl (3)}} \mid \boldsymbol{r}_i^{\text{rsa (3)}} \mid \boldsymbol{s}_i^{\text{searle (3)}} \mid \boldsymbol{t}_i^{\text{deontic (3)}} \right] \in \mathbb{R}^{30} \tag{1}$$

3 High-Fidelity Electromagnetic Transient (EMT) Systems and IEEE 39-Bus Testing

To ground the continuous-time physics layer into a verifiable industrial framework, sub-millisecond electromagnetic transient (EMT) numerical simulations are executed across a modified IEEE 39-bus test system via the open-source Julia platform `PowerSimulationsDynamics.jl`. Let the transient state of an electrical node i be characterized by a continuous differential-algebraic power vector $\boldsymbol{\xi}_i(t)$:

$$\boldsymbol{\xi}_i(t) = [V_i(t),\, \delta_i(t),\, P_{\text{IBR},i}(t),\, Q_{\text{IBR},i}(t)] \in \mathbb{R}^4 \tag{2}$$

where $V_i(t)$ represents the instantaneous voltage magnitude, $\delta_i(t)$ defines the phase angle deviation, and $P_{\text{IBR},i}(t), Q_{\text{IBR},i}(t)$ monitor the active and reactive power outputs of local inverter-based renewable resources.

4 In-Line Self-Adaptive Contraction-Constrained MPC Specifications

To guarantee asymptotic tracking stability under high-frequency stochastic phase noise obstructions, the secondary control loop implements an active, in-line Self-Adaptive Contraction-Constrained Model Predictive Control (MPC) law. The controller inserts a real-time system identification pass directly into the feedforward path, ingesting parameter estimate vectors $\theta = [\theta_{\text{phys}}, \theta_{\text{NN}}]^T$ recovered from partial measurements via a bi-level Moving Horizon Estimation (MHE) scheme following the paradigms of *Zhou & Köhler (2024)*.

Let the parameter-dependent state constraint space $\mathcal{X}(\theta_{\text{NN}})$ be dynamically morphed via a continuous in-line contraction multiplier $\alpha(\theta_{\text{NN}})$ bound against the nominal current saturation limit $\mathcal{X}_{\text{nominal}}$:

$$\mathcal{X}(\theta_{\text{NN}}) = \alpha(\theta_{\text{NN}}) \cdot \mathcal{X}_{\text{nominal}} = \left(\frac{1}{1 + \beta \cdot \theta_{\text{NN}}} \right) \cdot \mathcal{X}_{\text{nominal}} \tag{3}$$

where $\beta \in \mathbb{R}^+$ defines the invariant tuning gain. This constraint compression ensures that the active optimization trajectories contract exponentially toward a stable phase-locked manifold, avoiding optimization gradient collapse and preventing the physical inverter nodes from triggering control-loop interaction chaos.

4.1 Linear Matrix Inequality Formulation and Convergence Certification

Let $\boldsymbol{W}(x) = \boldsymbol{W}^T(x) > 0$ define the positive-definite dual contraction metric tensor. The system trajectories are certified stable under an L_2-gain noise attenuation index γ if the following symmetric matrix relation is strictly satisfied within the column boundaries:

$$\begin{bmatrix} \boldsymbol{\Theta}_{11}(x) & \boldsymbol{B}_w(x) \\ \boldsymbol{B}_w^T(x) & -\gamma^2 \boldsymbol{I} \end{bmatrix} \leq 0 \tag{4}$$

where the bounded sub-matrix component $\boldsymbol{\Theta}_{11}(x)$ is unrolled using a split block to prevent horizontal margin spills:

$$\begin{aligned} \boldsymbol{\Theta}_{11}(x) = &- \dot{\boldsymbol{W}}(x) + \boldsymbol{A}(x)\boldsymbol{W}(x) \\ &+ \boldsymbol{W}(x)\boldsymbol{A}^T(x) - 2\lambda \boldsymbol{W}(x) \end{aligned} \tag{5}$$

In this equation, $\boldsymbol{A}(x) = \frac{\partial f}{\partial x}$ acts as the system state Jacobian, $\boldsymbol{B}_w(x)$ maps the non-linear multiplicative phase noise vector parameters, and $\lambda \in \mathbb{R}^+$ defines the target exponential contraction convergence rate. This in-line formulation guarantees physical controllability before the state vectors can hit the unmodeled inner protective current rails.

5 Black-Hunter-Pan Multi-Ontology Dialogue Strategy

When independent corporate data centers must resolve grid access boundaries without data leakage or brute-force database merges, the deconfliction process is modeled as a preference-ordered dialogue game following the algebraic principles of **Black, Hunter & Pan**. Let $\mathcal{D}_t$ represent the sequential state-history matrix of the active dialogue at turn t. The turn-taking strategy function $\mathcal{S}_x(\mathcal{D}_t)$ acts as a deterministic, preference-ordered selection transducer over the Description Logic poset, mapped via the following strict dominance hierarchy:

$$\mathcal{S}_x(\mathcal{D}_t) = \text{Scounter}_x(\mathcal{D}_t) \succ \text{Sarg}_x(\mathcal{D}_t) \succ \text{Sconcede}_x(\mathcal{D}_t) \succ \langle x, \text{close}, \gamma \rangle \tag{6}$$

Transmission bandwidth overhead $\mathcal{C}$ scales as a compact function of decision forest graph depth d rather than the raw magnitude of the extensional grounding data:

$$\mathcal{C}(\mathcal{T}) \propto \mathcal{O}(d \cdot \Sigma_{\text{tokens}}) \ll \mathcal{C}(\mathcal{E}) \propto \mathcal{O}(|\mathcal{E}| \cdot \text{Dim}_{\text{features}}) \tag{7}$$

This enables the multi-agent perimeter to achieve absolute strategic consensus $\langle x, \text{close}, \gamma \rangle \cap \langle y, \text{close}, \gamma \rangle$ over thin network channels while keeping proprietary Level 1 telemetry perfectly firewalled.